Online Supplement

ClinicalTrials.gov preprocessing and auxiliary analyses

The final inclusion and exclusion criteria remain in Appendix C of the Supplementary Material. This document provides implementation details and additional clinical-trial plots.

Online Appendix A: Preprocessing Details

CTG Download. We downloaded the CTG results for completed studies with posted results as a JSON export (all available fields) from `https://clinicaltrials.gov/search?aggFilters=results:with,status:com` on August 7th, 2025. During data cleaning, we excluded four records with sample-size errors identified while reviewing candidate type I errors at $\alpha = .01$.

Record Construction. We transformed each ClinicalTrials.gov JSON record into flat, analysis-level rows prior to p-value cleaning and method harmonization. For each study (`nctId`), we iterated over outcome measures with at least one analysis and, for every outcome–analysis pair, emitted one row per denominator count with units "Participants" and with `groupId` in the analysis's declared `groupIds`. We carried forward study metadata (e.g., `briefTitle`, `studyType`), design-information fields (`interventionModel`, `observationalModel`, `timePerspective`, `allocation`), one-hot encoded phases (`EARLY_PHASE1`, `PHASE1`, `PHASE2`, `PHASE3`, `PHASE4`), and MeSH terms/IDs for conditions and interventions from the `derivedSection`. We also flagged whether the study included any references of type `RESULTS` or `DERIVED`.

Analysis Identifier. Each row was keyed by a stable, hash-based identifier,

$$\texttt{analysisId} = \mathrm{MD5}\big(\texttt{nctId} + \mathrm{JSON}(\texttt{outcome};\ \text{sorted keys}) + \mathrm{JSON}(\texttt{analysis};\ \text{sorted keys})\big),$$

which yields reproducible IDs for unique outcome–analysis pairs within a study (independent of JSON key order). This `analysisId` serves as the unit of aggregation in downstream preprocessing (e.g., computing sample sizes and $eJAB$).

Schema. Fields include: `analysisId`, `nctId`, `briefTitle`, `outcomeType`, `outcomeTitle`, `groupDescription`, `units`, `groupId`, `value`, `pValue`, `statisticalMethod`; MeSH fields (`condition_mesh`, `condition_ids`, `intervention_mesh`, `intervention_ids`); `hasResultsOrDerived`; study-design fields; and phase indicators. The field `value` records the number of participants for the given outcome group/arm.

Standardization of Reported p-Values. Reported p-values often included textual qualifiers (e.g., "<0.05"). We standardized them as follows:

1. Flag the presence of a "<" qualifier: `has_less_than` ← `grepl("<", pValue)`.
2. Extract the numeric component via regex:

$$\texttt{pValue} \leftarrow \texttt{str_extract(pValue, "[0-9]*.?[0-9]+")}.$$

3. Drop rows with missing or blank numeric components.
4. Coerce `pValue` to numeric and retain only valid probabilities: $0 <$ `pValue` < 1.
5. Exclude rows with zero sample size, which are incompatible with downstream model mappings.

Harmonization of Statistical Method Names. Free-text `statisticalMethod` labels were mapped to canonical test families using regular-expression rules (`str_detect`). Observations with no match were excluded. The exact mapping rules are:

Listing 1: Regex-based mapping of free-text methods to canonical families

```
statisticalMethod <- case_when(
  # --- T-TESTS ---
  str_detect(originalMethod, regex("\\bt[- ]?test", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("one[- ]?sample", ignore_case = TRUE)) ~ "1-ttest",

  str_detect(originalMethod, regex("\\bt[- ]?test", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("paired", ignore_case = TRUE)) ~ "1-ttest",

  str_detect(originalMethod, regex("\\bt[- ]?test", ignore_case = TRUE)) ~ "2-ttest",

  # --- CONDITIONAL LOGISTIC REGRESSION ---
  str_detect(originalMethod, regex("logistic", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("regression", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("conditional", ignore_case = TRUE)) ~ "clogit",

  # --- LOGISTIC REGRESSION ---
  str_detect(originalMethod, regex("logistic", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("regression", ignore_case = TRUE)) ~ "glm",

  # --- COX ---
  str_detect(originalMethod, regex("cox", ignore_case = TRUE)) &
    !str_detect(originalMethod, regex("wilcoxon|mantel|signed|rank", ignore_case = TRUE)) ~ "cox",

  # --- LOGRANK / MANTEL-COX ---
  str_detect(originalMethod, regex("log ?rank|mantel[- ]?cox", ignore_case = TRUE)) &
    !str_detect(originalMethod, regex("wilcoxon|signed", ignore_case = TRUE)) ~ "logrank",

  # --- MANN-WHITNEY ---
  str_detect(originalMethod, regex("mann[- ]?whitney", ignore_case = TRUE)) |
    (str_detect(originalMethod, regex("wilcoxon", ignore_case = TRUE)) &
       str_detect(originalMethod, regex("rank", ignore_case = TRUE)) &
       !str_detect(originalMethod, regex("signed", ignore_case = TRUE))) ~ "Utest",

  # --- WILCOXON SIGNED-RANK ---
  str_detect(originalMethod, regex("wilcoxon", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("signed", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("rank", ignore_case = TRUE)) ~ "wilcox",

  # --- KRUSKAL-WALLIS ---
  str_detect(originalMethod, regex("kruskal|wallis", ignore_case = TRUE)) ~ "Htest",

  # --- CHI-SQUARE ---
  str_detect(originalMethod, regex("chi[- ]?square|χ2|chi2|chisq", ignore_case = TRUE)) ~ "chisq",

  # --- REPEATED MEASURES ---
  str_detect(originalMethod, regex("repeated[- ]?measures?", ignore_case = TRUE)) ~ "rANOVA",

  # --- LINEAR REGRESSION ---
  str_detect(originalMethod, regex("linear", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("regression", ignore_case = TRUE)) ~ "lm",

  # --- ANOVA ---
  str_detect(originalMethod, regex("anova|analysis of variance", ignore_case = TRUE)) ~ "ANOVA",

  TRUE ~ NA_character_
)
```

Adjustment for One-Sided p-Values. For explicitly one-sided scalar analyses, we reconstructed the two-sided p-value as

$$p = 2\min\{p_1, 1 - p_1\}.$$

The null-centered one-sided eJAB omits $(n-1)/n$ from the exponent and uses the directional divisor $2(1-p_1)$. NHST instead requires $p_1 \leq \alpha/2$. Supplementary Appendix C.5 gives the method-label rules and full formula.

analysisId Sample Size QC. Certain tests should yield exactly one row per `analysisId` (e.g., one-sample t-test, Wilcoxon signed-rank test). To enforce consistency, we dropped analyses in which such tests appeared with multiple rows under the same `analysisId`.

Aggregation to the Test Level and Parameter Construction. We aggregated records at the `analysisId` (test) level. Fields invariant within a test were taken from the first row—e.g., `nctId`, `pValue`, `statisticalMethod`, `studyType`, phase indicators (`EARLY_PHASE1`, `PHASE1`, `PHASE2`, `PHASE3`, `PHASE4`), `outcomeTitle`, `outcomeType`, `groupDescription`, `conditionIds`, `briefTitle`, and `has_less_than`.

We defined the effective sample size n per test as:

$$n = \begin{cases} \texttt{first(value)}, & \text{one-sample } t\text{-test or Wilcoxon signed-rank;} \\ \max \texttt{value}, & \text{conditional logistic regression;} \\ 0.5 \sum \texttt{value}, & \text{Cox PH or log-rank;} \\ \sum \texttt{value}, & \text{otherwise.} \end{cases}$$

We also set method-specific parameters required by *eJAB*:

$$\texttt{I} = \begin{cases} n_{\text{rows within test}}, & \text{ANOVA, Kruskal–Wallis, repeated measures;} \\ \text{NA}, & \text{otherwise,} \end{cases}$$

$$(\texttt{R}, \texttt{C}) = \begin{cases} (n_{\text{rows within test}},\ 2), & \text{chi-squared test;} \\ (\text{NA},\ \text{NA}), & \text{otherwise.} \end{cases}$$

Mapping to JAB Models and Computation of eJAB. We mapped canonical `statisticalMethod` labels to the `JABO1` model strings required for $eJAB_{01}$. Mapping was performed only for tests included in our analysis (i.e., excluding chi-squared, Cox PH, log-rank, and repeated-measures designs):

Two-sample t-test, paired t-test, one-sample t-test → `"t-test"`,
Linear regression → `"linear_regression"`,
Logistic regression, conditional logistic regression → `"logistic_regression"`,
ANOVA → `"anova"`, Kruskal–Wallis → `"kruskal_wallis"`,
Mann–Whitney → `"mann_whitney"`, Wilcoxon → `"wilcoxon"`.

Given $(n, \texttt{pValue}, \texttt{model}, R, C, I)$ for each test, we computed $eJAB_{01}$ via

$$\texttt{JAB} = \texttt{JABO1}(n,\ \texttt{pValue},\ \texttt{model};\ R,\ C,\ I).$$

Exclusions and Final Dataset. We excluded rows with missing `model`, missing `JAB` or `pValue`, and tests with $n = 1$. The manuscript analysis then applied the criteria in Supplementary Appendix C to obtain the same screened set used throughout: 16,563 eligible primary- and secondary-outcome analyses from 1,790 exact Early Phase I, Phase I, and Phase II trials.

Online Appendix B: Auxiliary Clinical-Trial Figures

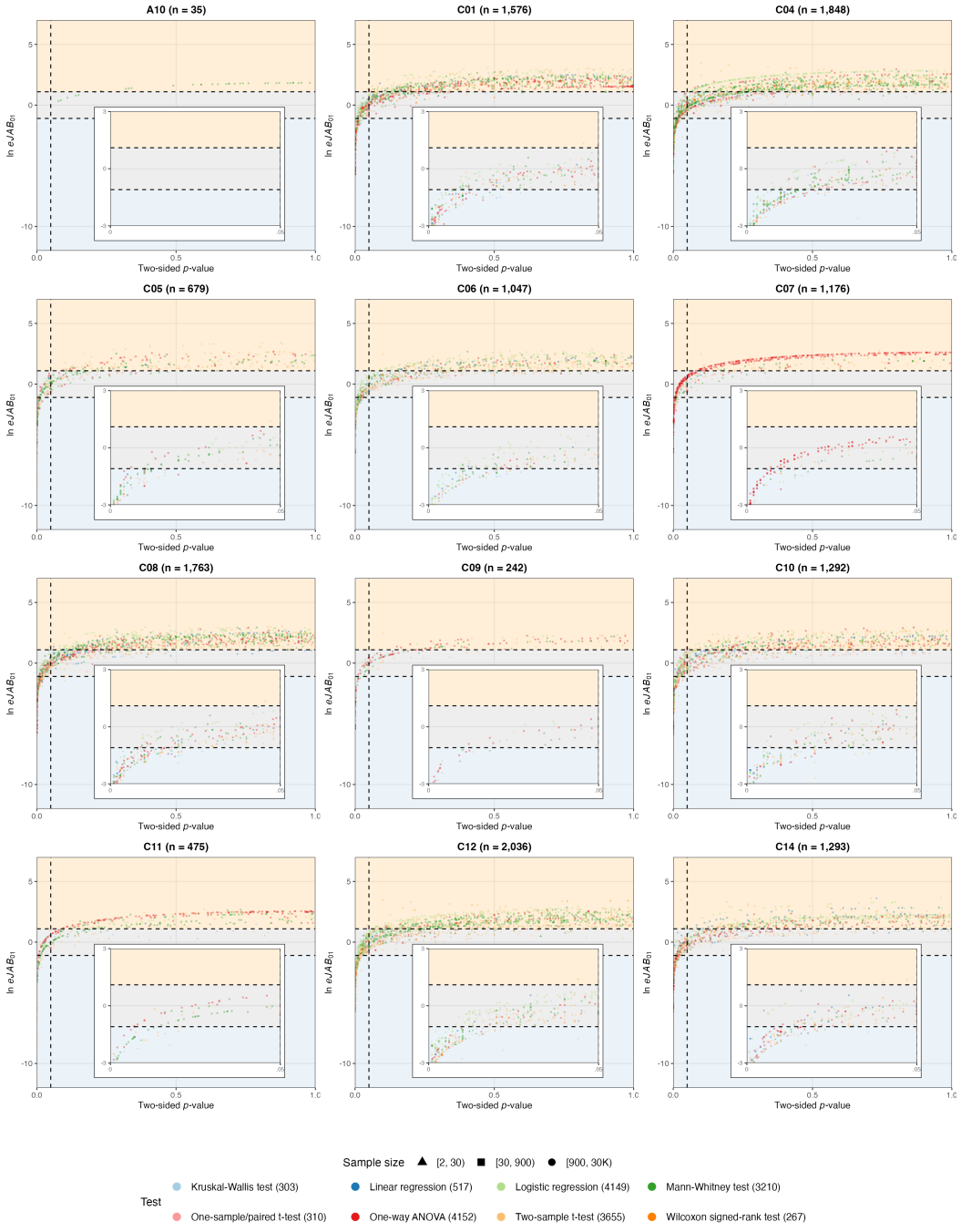


Figure O1: Scatterplots of $\ln eJAB_{01}$ versus two-sided p-values by MeSH condition (part 1 of 3), with $p = 2\min\{p_1, 1 - p_1\}$ for one-sided tests. Insets show $p \leq .05$. Colors identify tests, shapes sample-size groups, and shading evidence categories.

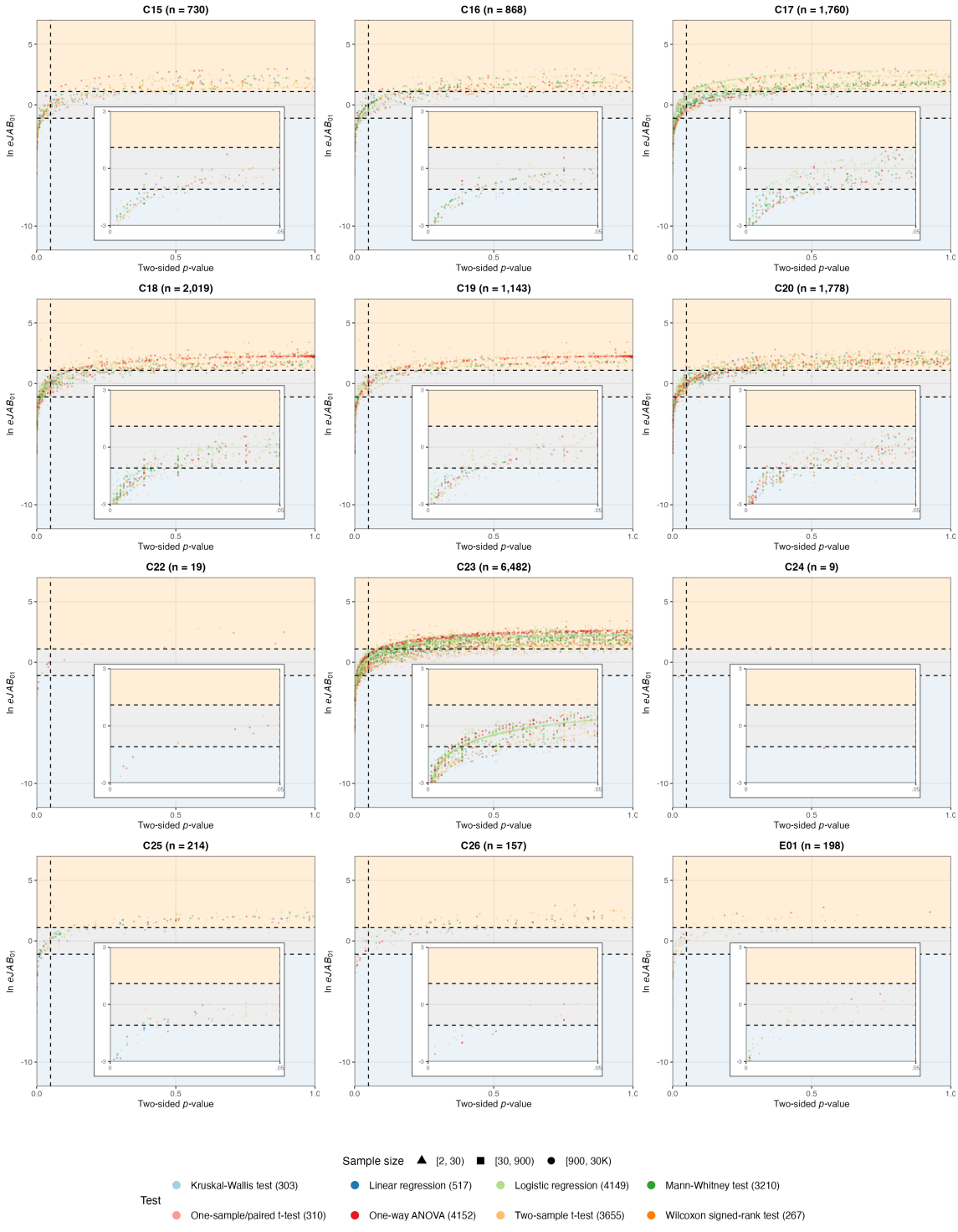


Figure O2: Scatterplots of $\ln eJAB_{01}$ versus two-sided p-values by MeSH condition (part 2 of 3), with $p = 2\min\{p_1, 1 - p_1\}$ for one-sided tests. Insets show $p \leq .05$. Colors identify tests, shapes sample-size groups, and shading evidence categories.

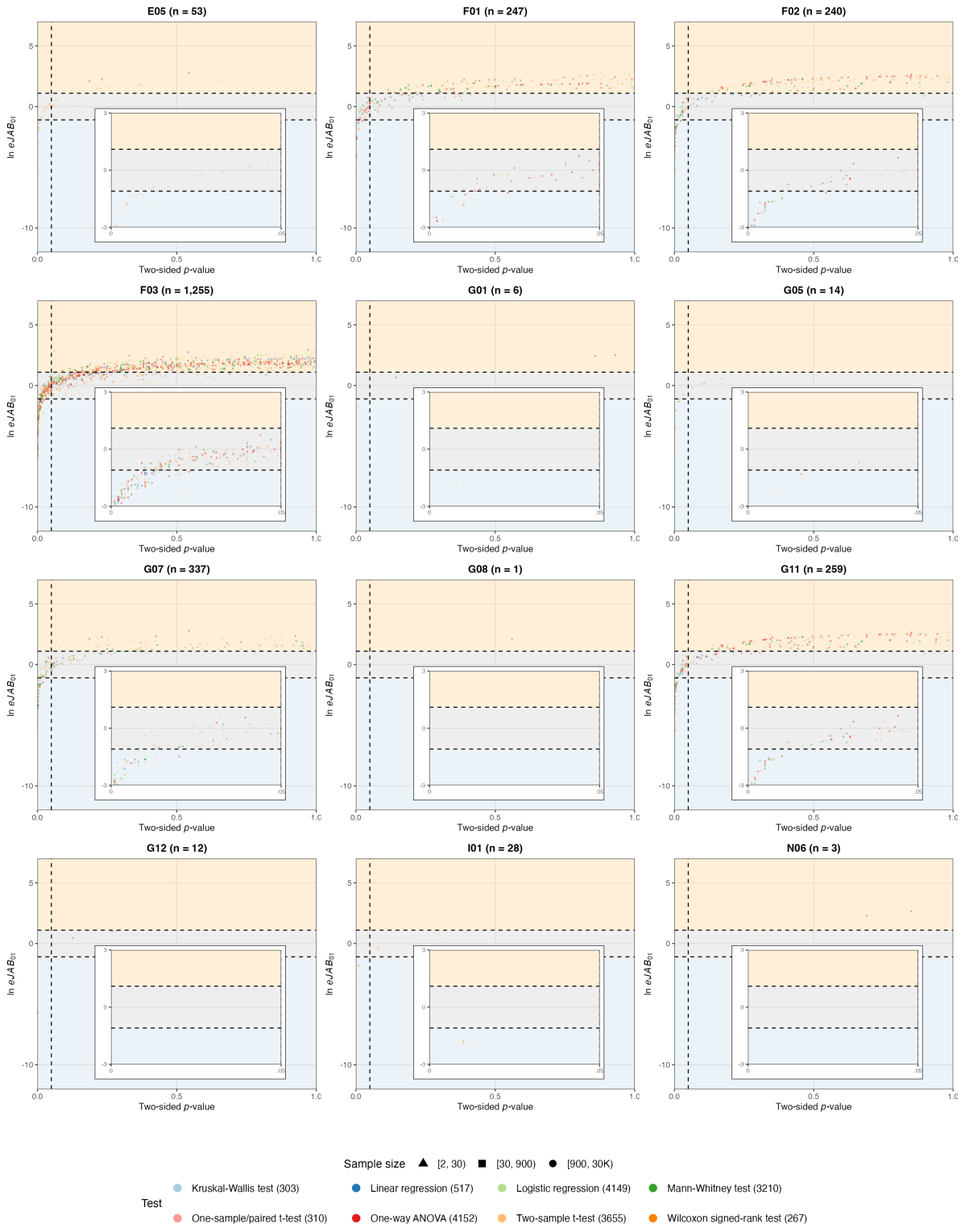


Figure O3: Scatterplots of $\ln eJAB_{01}$ versus two-sided p-values by MeSH condition (part 3 of 3), with $p = 2\min\{p_1, 1 - p_1\}$ for one-sided tests. Insets show $p \leq .05$. Colors identify tests, shapes sample-size groups, and shading evidence categories.

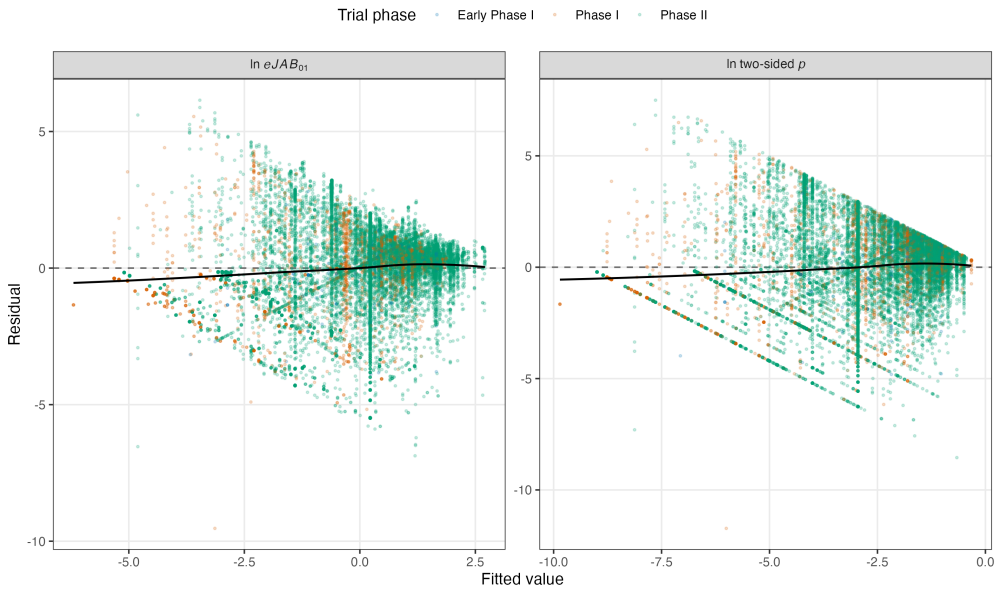


Figure O4: Residuals versus fitted values from random-intercept models for $\ln eJAB_{01}$ (left) and $\ln p$ (right), using $p = 2\min\{p_1, 1 - p_1\}$ for one-sided tests, in the screened primary- and secondary-outcome set. The dashed horizontal line marks zero residuals and the solid curve is a LOESS smooth. Nearly parallel slanted bands reflect heaping at common reported p thresholds.

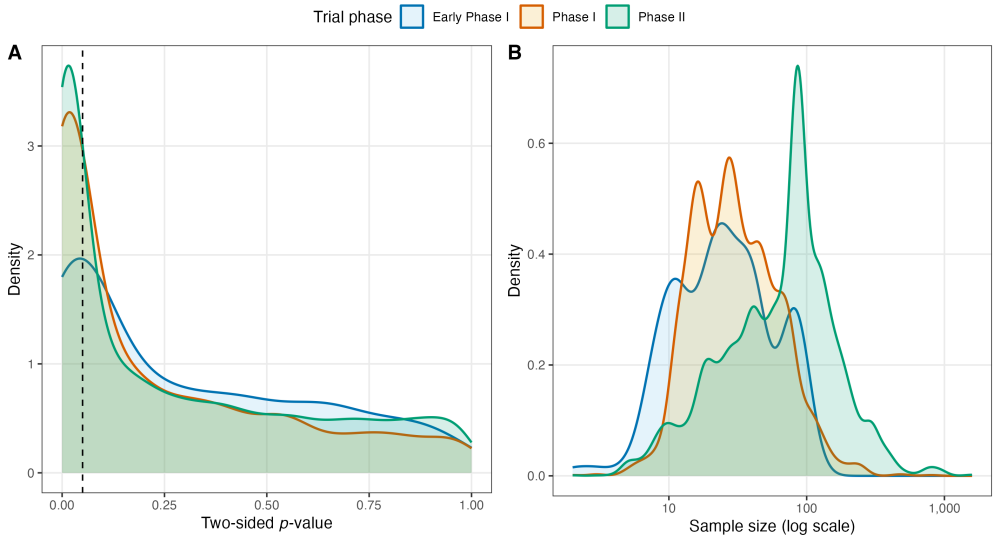


Figure O5: Distributions of two-sided p-values (A; vertical dashed line at .05), using $p = 2\min\{p_1, 1 - p_1\}$ for one-sided tests, and sample size (B) by exact Early Phase I, Phase I, and Phase II classification in the screened primary- and secondary-outcome set.